Maximizing the successful transmissions $\mathbf{1}_N^\top \mathbf{r}(\mathbf{x}, \mathbf{A})$ in a single slot is a maximum independent set problem. While repeating its solution across all slots maximizes the sum throughput, it serves none of the links outside the chosen set, as is the case for any stationary scheduling policy. To serve every link fairly, we impose per-link minimum transmission rate requirements $\boldsymbol{\delta} \in [0,1]^N$ on the long-term fraction of slots in which each link transmits successfully, and refer to the pair $(\mathbf{A}, \boldsymbol{\delta})$ as the *network state*.

Consider a *stochastic policy* $\mathcal{D}_{\mathbf{x}}(\cdot \mid \mathbf{A}, \boldsymbol{\delta})$, which is a conditional distribution over schedules from which $\mathbf{x}_t$ is drawn independently at each slot $t$. Throughout, we omit the conditioning on $(\mathbf{A}, \boldsymbol{\delta})$ when clear from context. An optimal policy $\mathcal{D}_{\mathbf{x}}^\star$ maximizes the expected sum throughput in (1) while satisfying the minimum transmission requirements over all distributions on $\{0,1\}^N$:

$$\begin{aligned} \mathcal{D}_{\mathbf{x}}^\star \in \underset{\mathcal{D}_{\mathbf{x}}}{\operatorname{argmax}} \quad & \mathbb{E}_{\mathcal{D}_{\mathbf{x}}}\big[\mathbf{1}_N^\top \mathbf{r}(\mathbf{x}, \mathbf{A})\big] \\ \text{subject to} \quad & \mathbb{E}_{\mathcal{D}_{\mathbf{x}}}\big[\mathbf{r}(\mathbf{x}, \mathbf{A})\big] \geq \boldsymbol{\delta}. \end{aligned} \tag{2}$$

Since at most one link in any clique of the conflict graph may transmit in a given slot, an optimal policy $\mathcal{D}_{\mathbf{x}}^\star$ for (2) samples independent sets. As such, these solution distributions are highly multimodal, and solving for them over $2^N$ schedules is computationally prohibitive for varying large-scale network deployments.

We circumvent this by introducing a parameterization through $\boldsymbol{\theta}$ and learning a parameterized policy $\mathcal{D}_{\mathbf{x}}(\boldsymbol{\theta})$. This shifts our goal to finding the optimal parameterized conditional distribution $\mathcal{D}_{\mathbf{x}}(\boldsymbol{\theta}^\star)$ that approximates the target stochastic policies,

$$\boldsymbol{\theta}^\star \in \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\ \mathbb{E}_{(\mathbf{A},\boldsymbol{\delta})\sim\mathcal{D}_{\mathbf{A},\boldsymbol{\delta}}}\big[D_{\mathrm{KL}}\big(\mathcal{D}_{\mathbf{x}}^\star \,\|\, \mathcal{D}_{\mathbf{x}}(\boldsymbol{\theta})\big)\big], \tag{3}$$

where $\mathcal{D}_{\mathbf{A},\boldsymbol{\delta}}$ denotes the training distribution of network states. Solving (3) requires only samples from $\mathcal{D}_{\mathbf{x}}^\star$, and not its closed form. We generate the samples as trajectories of a dual descent expert and replace $\mathcal{D}_{\mathbf{x}}^\star$ with their empirical distribution $\widehat{\mathcal{D}}_{\mathbf{x}}$ (Section 2.1).

We treat schedules as graph signals on the conflict graph and parameterize $\mathcal{D}_{\mathbf{x}}(\boldsymbol{\theta})$ as a diffusion model conditioned on the network state $(\mathbf{A}, \boldsymbol{\delta})$, from which we sample scheduling decisions $\mathbf{x} \sim \mathcal{D}_{\mathbf{x}}(\boldsymbol{\theta}^\star)$ (Section 3).

### 2.1. Dual Gradient Descent

The expert solves (2) in the dual domain, where the search over distributions reduces to a sequence of problems over single schedules. Introducing nonnegative multipliers $\boldsymbol{\lambda} \in \mathbb{R}_+^N$, with one multiplier for each link's minimum transmission constraint, the corresponding Lagrangian for (2) is

$$\mathcal{L}(\mathcal{D}_{\mathbf{x}}, \boldsymbol{\lambda}) = \mathbf{1}_N^\top \mathbb{E}_{\mathcal{D}_{\mathbf{x}}}\big[\mathbf{r}(\mathbf{x}, \mathbf{A})\big] + \boldsymbol{\lambda}^\top \big(\mathbb{E}_{\mathcal{D}_{\mathbf{x}}}\big[\mathbf{r}(\mathbf{x}, \mathbf{A})\big] - \boldsymbol{\delta}\big). \tag{4}$$

We approximate the expectations in (4) using empirical averages over a finite horizon of $T$ scheduling steps. The long-term Lagrangian can be reduced to an instantaneous formulation [8]. Let $\boldsymbol{\lambda}_t \in \mathbb{R}_+^N$ denote the vector of Lagrange multipliers at step $t \in [T]$. The corresponding *instantaneous Lagrangian* is

$$\begin{aligned} \mathcal{L}_I(\mathbf{x}, \boldsymbol{\lambda}_t) &= \mathbf{1}_N^\top \mathbf{r}(\mathbf{x}, \mathbf{A}) + \boldsymbol{\lambda}_t^\top \big(\mathbf{r}(\mathbf{x}, \mathbf{A}) - \boldsymbol{\delta}\big) \\ &= (\mathbf{1}_N + \boldsymbol{\lambda}_t)^\top \mathbf{r}(\mathbf{x}, \mathbf{A}) - \boldsymbol{\lambda}_t^\top \boldsymbol{\delta}. \end{aligned} \tag{5}$$

At each scheduling step $t$, we select a schedule $\mathbf{x}_t$ that maximizes the instantaneous Lagrangian for the current multipliers $\boldsymbol{\lambda}_t$:

$$\mathbf{x}_t \in \underset{\mathbf{x}}{\operatorname{argmax}}\ \mathcal{L}_I(\mathbf{x}, \boldsymbol{\lambda}_t). \tag{6}$$

We then apply gradient descent to minimize the dual function, $g(\boldsymbol{\lambda}_t) = \max_{\mathbf{x}} \mathcal{L}_I(\mathbf{x}, \boldsymbol{\lambda}_t)$, with respect to the multipliers:

$$\boldsymbol{\lambda}_{t+1} = [\boldsymbol{\lambda}_t - \eta(\mathbf{r}(\mathbf{x}_t, \mathbf{A}) - \boldsymbol{\delta})]_+. \tag{7}$$

Here, $\eta > 0$ denotes the step size, and the rate residual $\mathbf{r}(\mathbf{x}_t, \mathbf{A}) - \boldsymbol{\delta}$ is a gradient of the dual function at $\boldsymbol{\lambda}_t$.

Since $\boldsymbol{\lambda}_t^\top \boldsymbol{\delta}$ does not depend on $\mathbf{x}$, maximizing (5) is equivalent to maximizing $(\mathbf{1}_N + \boldsymbol{\lambda}_t)^\top \mathbf{r}(\mathbf{x}, \mathbf{A})$. Since $\mathbf{r}(\mathbf{x}, \mathbf{A})$ always indicates an independent set, this maximization can be restricted without loss of optimality to the set $\mathcal{I} \subseteq \{0,1\}^N$ of schedules whose active links form an independent set in the conflict graph, for which $\mathbf{r}(\mathbf{x}, \mathbf{A}) = \mathbf{x}$. Starting from $\boldsymbol{\lambda}_1 = \mathbf{0}$, each step thus solves a maximum weighted independent set (MWIS) problem,

$$\mathbf{x}_t \in \underset{\mathbf{x}\in\mathcal{I}}{\operatorname{argmax}}\ \mathbf{w}_t^\top \mathbf{x}, \qquad \mathbf{w}_t = \mathbf{1}_N + \boldsymbol{\lambda}_t. \tag{8}$$

We then update the multipliers using (7), which determines the weights for the next step. Repeating this procedure for $T$ steps yields the expert sequences $\mathbf{x}_{1:T} \sim \widehat{\mathcal{D}}_{\mathbf{x}}$. Since solving an MWIS problem at every step is costly (Figure 2(c)), we learn the empirical distribution with a diffusion model.

## 3. GRAPH-CONDITIONED DIFFUSION MODEL

A scheduling decision $\mathbf{x}$ assigns one value to each link and is thus a signal on the nodes of the conflict graph $\mathbf{A}$. We parameterize the learned policy $\mathcal{D}_{\mathbf{x}}(\boldsymbol{\theta})$ as a diffusion model of such graph signals [17, 21], conditioned on the network state $(\mathbf{A}, \boldsymbol{\delta})$. We distinguish the continuous diffusion variable from the binary schedule and let $\mathbf{z}_0 \in \mathbb{R}^N$ denote an expert schedule for given $(\mathbf{A}, \boldsymbol{\delta})$, i.e., $\mathbf{z}_0 \sim \widehat{\mathcal{D}}_{\mathbf{x}}$.

The forward diffusion process corrupts $\mathbf{z}_0$ with Gaussian noise over $M$ steps. The noisy signal at any step $m$ is given by

$$\mathbf{z}_m = \sqrt{\bar{\alpha}_m}\, \mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_m}\, \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N), \tag{9}$$

where $\bar{\alpha}_m = \prod_{i=1}^m \alpha_i$ and $\alpha_m = 1 - \beta_m$ constitute the noise schedule chosen specifically so that $\mathbf{z}_M \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$ regardless of $\mathbf{z}_0$.

We learn a noise predictor $\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{z}_m, m; \mathbf{A}, \boldsymbol{\delta})$ that undoes this corruption by removing the noise $\boldsymbol{\epsilon}$ from $\mathbf{z}_m$, $m$ and network state $(\mathbf{A}, \boldsymbol{\delta})$. We train it to minimize the DDPM objective [21],

$$\boldsymbol{\theta}^\star \in \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\ \mathbb{E}\Big[\big\|\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{z}_m, m; \mathbf{A}, \boldsymbol{\delta}) - \boldsymbol{\epsilon}\big\|_2^2\Big], \tag{10}$$

where the expectation is over the training network states $(\mathbf{A}, \boldsymbol{\delta})$ and noisy samples $\mathbf{z}_m$ formed via (9) from independent draws of expert schedules $\mathbf{z}_0 \sim \widehat{\mathcal{D}}_{\mathbf{x}}(\cdot \mid \mathbf{A}, \boldsymbol{\delta})$, diffusion step $m \sim \mathrm{Unif}\{1, \ldots, M\}$, and noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N)$. We note that the objective in (10) is a variational bound on the negative log-likelihood of the expert samples [21], and thus a tractable surrogate for (3).

At inference, we sample from the trained model by drawing $\mathbf{z}_M \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N)$ and running a deterministic DDIM sampler [22]. Specifically, we estimate the clean sample at each step $m$ as

$$\hat{\mathbf{z}}_0 = \frac{\mathbf{z}_m - \sqrt{1 - \bar{\alpha}_m}\, \boldsymbol{\epsilon}_{\boldsymbol{\theta}^\star}(\mathbf{z}_m, m; \mathbf{A}, \boldsymbol{\delta})}{\sqrt{\bar{\alpha}_m}}, \tag{11}$$

and use this estimate to obtain the next iterate as

$$\mathbf{z}_{m-1} = \sqrt{\bar{\alpha}_{m-1}}\, \hat{\mathbf{z}}_0 + \sqrt{1 - \bar{\alpha}_{m-1}}\, \boldsymbol{\epsilon}_{\boldsymbol{\theta}^\star}(\mathbf{z}_m, m; \mathbf{A}, \boldsymbol{\delta}), \tag{12}$$

for $m = M, \ldots, 1$ with $\bar{\alpha}_0 = 1$. A binary schedule $\mathbf{x}_t$ at each slot $t$ is obtained by thresholding the entries of the final iterate $\mathbf{z}_0$ at 1/2.